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18 July 1984
SUPERSEDING
MIL-C-62006(MO)
2 June 1964

MILITARY SPECIFICATION

COOLER, OIL: TRANSMISSION AND ENGINE

This specification is approved for use by the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers aluminum oil cooler assemblies for air cooling engine and transmission oil (see 6.1) and for the individual coolers and cores which make up the assembly types as specified (see 6.2):

TYPE I	- Engine and transmission oil cooler ends, part numbers 7003291, 7003292, 7985070
TYPE II	- Engine and transmission cores, side, part numbers 7003293, 7003294
TYPE III	- Engine and transmission oil cooler assemblies, part numbers 7003290, 8698621

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: DRSTA-GSS, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 2930

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS
MILITARY

MIL-L-2104	- Lubricating Oil, Internal Combustion Engine Heavy Duty.

STANDARDS
MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-193	- Painting Procedures, Tactical Vehicles (Tracked and Wheeled).
MIL-STD-45662	- Calibration Systems Requirements.

2.1.2 Other Government documents, drawings, and publications. The following documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS
MILITARY

7003290	- Coolers Assy, Engine and Transmission Oil.
7003291	- Core, Oil Cooler, End.
7003292	- Core, Oil Cooler, End.
7003293	- Core, Oil Cooler, Side.
7003294	- Core, Oil Cooler, Side.
7985070	- Core, Oil Cooler.
8698621	- Cooler Assy, Engine and Transmission Oil.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1. First article. Unless otherwise specified, the contractor shall furnish cooler assemblies, coolers or cores which shall be subjected to first article inspection (see 4.4). First article inspection samples, properly marked with identifying information shall be representative of the unit to be furnished to the Government. All subsequent cooler assemblies, coolers or cores delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Materials. Materials shall be as specified on applicable standards, drawings, and specifications (see 6.5). Materials shall be free from defects which adversely effect performance or serviceability of the finished product.

3.3 Construction. Cooler assembly, coolers and core physical and functional interfaces shall conform to Drawings 7003290, 7003291, 7003292, 7003293, 7003294, 7985070, or 8698621 as applicable (see 6.2).

3.3.1. Flux control. All cooler assemblies, coolers and cores shall be free of flux after performance of the welding operation. Each unit shall be filled with water at 180 to 200 degrees Fahrenheit (°F) and circulated through the core for 10 minutes. The chloride content shall not exceed 40 parts per million above existing level for local water.

3.3.2 Contamination resistance. All cooler assemblies, coolers and cores shall be free of contamination. Each unit shall be half-filled with perchlorethylene or trichlorethylene, capped and shaken in both horizontal and vertical planes. Drain both parts of the unit through Whitman No. 41, or equivalent filter paper. There shall be no visible particles on the filter paper after drainage is completed.

3.3.3 Interchangeability. All parts of the oil cooler assemblies shall be interchangeable with parts of other manufacturers having the same military part number.

3.3.4 Exterior surface treatment. All exposed exterior surfaces of the cores shall be cleaned, painted, or treated for corrosion resistance as specified on the applicable drawings, or if not so specified, in accordance with applicable provisions of MIL-STD-193.

3.4 Performance characteristics.

3.4.1 Leakage. Cooler assemblies, coolers or cores shall not leak when subjected to the following pressures:

3.4.1.1 Pneumatic pressure. Cooler assemblies, coolers or cores shall not leak when subjected to a pneumatic pressure of 100 pounds per square inch (psi), when submerged in water at 120 to 140°F (see 4.6.1.1).

3.4.1.2 Hydrostatic pressure. Coolers shall show no evidence of leakage when subjected to a hydrostatic dry air or nitrogen pressure of 400 psi (see 4.6.1.2).

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3.4.1.3 Hydraulic pressure. Cores shall show no evidence of leakage when subjected to 15,000 hydraulic pressure cycles from 0 to 300 pounds per square inch (psi) to 0 at the rate of 13 to 20 cycles per minute using 50 grade oil, that conforms to MIL-L-2104, at a temperature of 170 to 200°F (see 4.6.1.3).

3.4.2 Heat rejection. Cooler assemblies and cores shall have the heat rejection capacity for the type unit specified:

3.4.2.1 Type I cores. Heat rejection for type I engine and transmission end cores shall be not less than BTU per minute indicated (see 4.6.2.1):

a. 2729 BTU - Engine (P/N 7003292)
b. 6484 BTU - Transmission (P/N 7003291 and 7985070)

3.4.2.2 Type II cores. Heat rejection for type II engine and transmission side cores shall be not less than BTU per minute indicated (see 4.6.2.2).

a. 1354 BTU - Engine (P/N 7003293)
b. 1643 BTU - Transmission (P/N 7003293)
c. 1266 BTU - Engine (P/N 7003294)
d. 1563 BTU - Transmission (P/N 7003294)

3.4.2.3 Type III cooler assembly. Heat rejection for type III, engine and transmission cooler assemblies (P/N 7003290 and 8698621) shall be not less than BTU per minute indicated (see 4.6.2.3).

a. 5280 BTU - Engine
b. 9525 BTU - Transmission

3.5 Identification and marking. Unless otherwise specified, identification and marking of the cooler assemblies or cores shall be in accordance with MIL-STD-130.

3.6 Workmanship. Workmanship shall be of the quality necessary to produce cooler assemblies and cores free from defects which may affect their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Inspection equipment. Unless otherwise specified in the contract, the supplier is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10 percent of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.1.2 Materials and construction. To determine conformance to 3.2 through 3.3.4, inspection and material certification records shall be maintained by the contractor. Records shall be subject to review by the Government and shall include date, part, or characteristic identification, inspection results, and disposition of lot (accepted or rejected). Corrective action taken on noted defects shall be subject to approval by the Government.

4.2 Classification of inspection:

a. First article inspection (see 4.4).
b. Quality conformance inspections (see 4.5).
   1). Examination (see 4.5.2).
   2). Acceptance tests (see 4.5.3.1).
   3). Control tests (see 4.5.3.2).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be conducted under the following conditions:

a. Air temperature 75° $\pm$ 10°F
b. Barometric pressure $\overline{28}$.54 $^{+2}_{-3}$ inches mercury (Hg)
c. Relative humidity 50 $\pm$ 30 percent

4.3.1 Calibration of test equipment. Unless otherwise specified, test equipment accuracy shall be capable of measurement of 10 percent of product or test tolerance. Calibration of test equipment shall be conducted at intervals which serve to establish required accuracy. Records of calibration shall be made available to the Government. The Government inspector shall refuse to allow inspection where accuracy of test equipment has not been established to the satisfaction of the Government.

4.4 First article inspection. Unless otherwise specified (see 6.2), the Government shall select three cooler assemblies or cores produced under the production contract (see 6.2) for first article inspection. First article samples shall be examined as specified in 4.5.2 and tested as specified in table II. Approval of the first article sample by the Government shall not relieve the contractor of his obligation to supply cooler assemblies or cores that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

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4.4.1 Initial production inspection. Unless otherwise specified (see 6.2), the Government shall select 2 assemblies from the first 100 cooler assemblies or cores produced under the production contract for initial production inspection. Initial production units shall be examined as specified in 4.5.2 and tested as specified in table II.

4.4.2 First article inspection failure. Failure of any first article sample to pass specified examinations or tests shall be cause for refusal to grant first article approval until corrective action by the contractor has been approved by the Government.

TABLE I. Classification of defects.

Category	Defect	Method of examination
Critical	None defined	
Major	AQL 1.0 percent	
101	Dimensions affecting interchangeability, out of tolerance.	Visual/ Gage
102	Leakage, oil (external).	Functional
Minor	AQL 2.5 percent	
201	Identification marking improper.	Visual
202	Dimensions not affecting interchangeability, out of tolerance.	Visual/ Gage
203	Painted surfaces, finish peeling, blistered or scaling.	Visual
204	Workmanship, faulty.	Visual

TABLE II. Classification of tests.

Title	Requirement	Test	First article	Quality conformance	
				Acceptance (100%)	Control
Leakage					
Pneumatic pressure	3.4.1.1	4.6.1.1	X	X	
Hydrostatic pressure	3.4.1.2	4.6.1.2	X		
Hydraulic pressure	3.4.1.3	4.6.1.3	X		
Heat rejection					
Type I cores	3.4.2.1	4.6.2.1	X		X
Type II cores	3.4.2.2	4.6.2.2	X		X
Type III cores	3.4.2.3	4.6.2.3	X		X

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4.5 Quality conformance inspection.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all the cooler assemblies, coolers or cores of one type from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105.

4.5.2 Quality conformance examinations. Cooler assemblies or cores selected in accordance with 4.5.1.2 shall be examined for conformance to applicable drawings and this specification (3.3). Examinations shall be performed against the classification of defects and with the acceptable quality levels (AQL's) specified in 4.5.2.1 and table I.

4.5.2.1 Acceptable quality level. Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following acceptable quality levels (AQL) on the basis of percent defective (see 6.3).

Classification	AQL
Major	1.0
Minor	2.5

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as listed in table I.

4.5.3 Quality conformance tests.

4.5.3.1 Acceptance tests (100 percent). Each cooler assembly or core shall be subjected to the tests specified in table II.

4.5.3.2 Control tests. Control tests shall be conducted on one cooler assembly, cooler or core from each lot of 200 units consecutively produced, except that not more than one test shall be performed in a three month period, not less than one test in a six month period. The cooler assemblies, coolers or cores shall be subjected to the tests specified in table II.

4.5.4 Failure. Failure of any cooler assembly, cooler or core to pass any of the specified tests shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.6 Methods of inspection.

4.6.1 Leakage tests.

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4.6.1.1 Pneumatic pressure test. To determine conformance to 3.4.1.1, each cooler assembly, cooler or core shall be submerged in water at a temperature of 120 to 140°F. While submerged the internal pressure shall be gradually applied, using dry air or nitrogen, until a pressure of 100 psi has been reached. The pressure shall be maintained for a period of 1 minute.

4.6.1.2 Hydrostatic pressure test. To determine conformance to 3.4.1.2 each cooler assembly shall have a pressure of 400 psi applied using dry air or nitrogen hydrostatically and maintained for a period of 10 minutes.

4.6.1.3 Hydraulic pressure test. To determine conformance to 3.4.1.3, if specified (see 6.2) cores selected at random shall be subjected to 15,000 hydraulic pressure cycles on a test stand capable of maintaining the specified pressure, cycle rate and temperature. This is a destructive test, so test units shall be indelibly marked "Do not use" and discarded.

4.6.2 Heat rejection tests. To determine conformance to 3.4.2, heat rejection tests shall be conducted using oil of the grade indicated entering the cooler assemblies, coolers or cores at the oil flow rates, air inlet and outlet temperatures and air flow rates listed as follows:

4.6.2.1 Type I cores. To determine conformance to 3.4.2.1, cores shall be subjected to the following tests:

		Engine	Transmission
a.	Air		
(1)	Flow, lbs/min	170	275
(2)	Static drop inches H_2O	11	7.5
(3)	Inlet temperature °F	140	140
b.	Oil		
(1)	Grade (per MIL-L-2104)	15W-40	10
(2)	Flow gpm	150 lbs/min	284 gpm
(3)	Pressure	30	30
(4)	Inlet temperature, °F	-	300
(5)	Outlet temperature, °F	225	-

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4.6.2.2 Type II cores. To determine conformance to 3.4.2.2, cores shall be subjected to the following tests:

	Engine	Transmission	Engine	Transmission
a. Air				
(1) Flow, lbs/min	89	69	84	65
(2) Static drop inches H_2O	11	7.5	11	7.5
(3) Inlet temperature, °F	140	140	140	140
b. Oil				
(1) Grade (per MIL-L-2104)	15W-40	10	15W-40	10
(2) Flow gpm	80	78	80	78
(3) Pressure drop psi	30	25	30	25
(4) Inlet temperature, °F	-	300	-	300
(5) Outlet temperature, °F	225	-	225	-

4.6.2.3 Type III cooler assemblies. To determine conformance to 3.4.2.3, cooler assemblies shall be subjected to the following tests:

	Engine	Transmission
a. Air		
(1) Flow, lbs/min	900	900
(2) Static drop inches H_2O	11	7.5
(3) Inlet temperature, °F	140	140
b. Oil		
(1) Grade (per MIL-L-2104)	15W-40	10
(2) Flow gpm	310	440
(3) Pressure drop psi	30	25
(4) Inlet temperature, °F	-	300
(5) Outlet temperature, °F	225	-

4.7 Identification and marking. The cooler assemblies, coolers or cores shall be visually examined to verify conformance to the identification and marking requirements of 3.5.

4.8 Workmanship. Workmanship examinations shall be performed phases of the assembly process to verify conformance with 3.6.

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5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging standard or packaging data sheet specified by the contracting authority (see 6.2).

6. NOTES

6.1 Intended use. The cooler assemblies or cores covered by this specification are intended for use in military tactical and transport vehicles to cool the engine and transmission oil.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. If preproduction samples shall be furnished (see 3.1).
c. If initial production units shall not be furnished (see 3.1).
d. Selection of applicable level and packaging standard or packaging data sheet (see 5.1).

6.3 Process average. Sampling may be initiated if the process average value for the first twenty cooler assemblies or cores inspected is less than the AQL specified in 4.5.2.1 for major and minor defects.

$$\text{Process average} = \frac{\text{Number of defects}}{\text{Number of cooler assemblies or cores}} \times 100$$

If the computed process average exceeds the specified AQL, 100 percent inspection shall be performed and continued until such time that the process average for twenty consecutive cooler assemblies or cores is less than the specified AQL (see 4.5.2.2).

6.4 Disposition of test assemblies. Assemblies undergoing destructive tests as 4.6.1.3 shall not be used and shall be indelibly marked "DO NOT USE", and purposely scrapped.

6.5 Recycled materials. The use of recycled materials which meet the requirements of the applicable material specifications without jeopardizing the intended use of the item shall be encouraged (see 3.2).

6.6 Cross-reference. Type I (transmission) and II (engine) oil coolers as listed in the previous issue are obsolete, and they were deleted from this revision. Type III, IV and V, as listed in the previous issue, were renumbered Type I, II and III respectively.

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6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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